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Simulation Sciences Seminar Project Report

Benchmarking Sampling Methods for Probabilistic Calibration in Mass Flow Model

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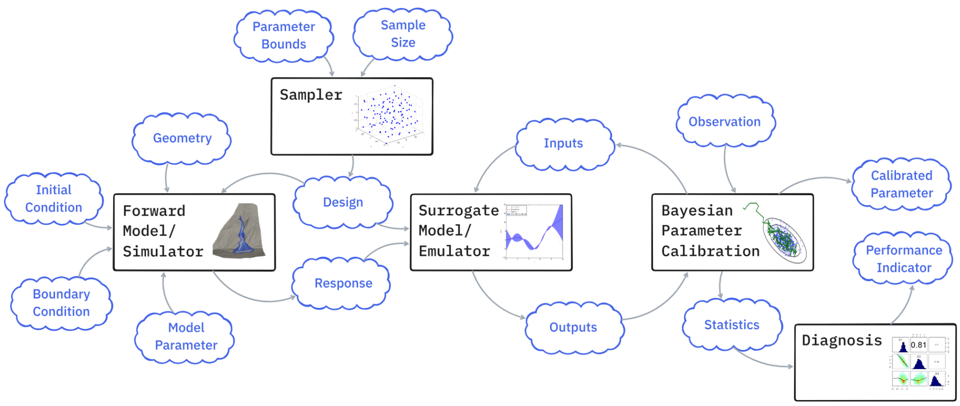

Benchmarking Sampling Methods for Probabilistic Calibration in Mass Flow Model

Enes Çağlar Korkmazgöz

Abstract

Here I will concentrate on the following:

Context: Bayesian calibration of the Voellmy mass flow model (friction parameters mu, xi) via an R-based RobustGaSP surrogate served through UM-Bridge.

Motivation/Gap: ADD!!!!!!!

Approach: Benchmark number of samplers (currently we have three, emcee, rwmcmc, dynesty) against the same sampler-agnostic posterior in a Nextflow pipeline.

Results: These will be key results in short. All samplers are expected to converge to consistent posteriors, but they will differ in computational cost, implementation complexities, and other parameters we described in the presentation.

Takeaway: Practical guidance on the sampler choice for calibration problems.

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1 Introduction

I will just make the framing here. Parameter uncertainty is a key problem in many scientific and engineering applications. In this report, I will focus on the Bayesian calibration of the Voellmy mass flow model via an R-based RobustGaSP surrogate served through UM-Bridge (I do not think I even need that explanations but since that is a report, it might be better to mention them). The main goal is to benchmark a number of samplers against the same sampler-agnostic posterior in a Nextflow pipeline. The samplers we will consider are emcee, rwmcmc, dynesty...

1.1 Motivation

Motivation (broad problem -> Bayesian -> sampling -> benchmarking)

Consider a common situation in computational science: we have a physical model of a process, and we have a set of measurements of that process. The model contains parameters that cannot be measured directly and must instead be chosen such that the model output matches the observations. In this work, the process is a mass flow event described by the Voellmy friction model, and the unknown parameters are the friction coefficients μ and ξ.

[Here, I want to insert a graphical illustration]

A natural first attempt is deterministic: search for the single parameter combination that fits the data best, for example by minimizing the misfit between model output and measurements. In practice, this approach runs into three difficulties. First, the measurements are noisy, so a parameter set that reproduces the data exactly is partly fitting measurement error rather than the underlying physics. Second, the problem is often ill-posed: distinct parameter combinations can produce nearly indistinguishable model outputs, so the "best" point returned by an optimizer is only one of many almost equally plausible answers, and the method gives no indication that the others exist. Third, the model itself is imperfect, since it is a simplified description of reality, and in our case it is additionally replaced by a surrogate for computational reasons, so no single parameter value can be regarded as the true one. A single number therefore answers the wrong question. The question we actually need to answer is: *which parameter values are consistent with the observations, and how plausible is each of them?*

This question calls for a distribution over parameters rather than a point estimate, and this is precisely what the Bayesian approach provides. Bayesian calibration treats the parameters as random variables: prior knowledge, such as physically admissible ranges, is encoded in a prior distribution, and the measurements update this prior into a posterior distribution via the likelihood. The posterior captures everything the data can tell us about the parameters, including how uncertain we remain and how parameters trade off against each other. This uncertainty can then be propagated to model predictions, which matters whenever the model informs real decisions, as is the case for mass flow hazard assessment.

The posterior is known only up to a normalizing constant and has no closed form, so it can only be explored by methods that evaluate it point by point - MCMC methods and their relatives are the standard tools for this task.

$$p(\theta \mid y) = \text{an unknown distribution}$$

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$$

Yet "use a sampler" is not the end of the story, because samplers differ substantially in how they explore the posterior. Random-walk Metropolis-Hastings is simple but requires manual tuning; ensemble methods such as emcee reduce the tuning burden; nested sampling, as implemented in dynesty, follows a different principle altogether and additionally estimates the model evidence (I will add other samplers here). These methods differ in convergence behavior, which is exactly the situation in this work, where the model is an R-based Gaussian process surrogate accessed through UM-Bridge. The choice of sampler is therefore consequential but far from obvious. This report addresses that gap by benchmarking many different samplers on the same calibration problem, with an identical posterior definition and an identical pipeline, so that the observed differences can be attributed to the samplers themselves. [1]

[1] I can give some resources here, like the papers of packages and "Handbook of Monte Carlo Methods"

1.2 Research Questions

Our research questions are as follows:

RQ1 Do emcee, rwmcmc, dynesty, and other samplers converge to consistent posterior estimates for the Voellmy friction parameters μ and ξ?

RQ2 How do the all samplers compare in computational cost?

RQ3 Which implementation challenges arise when integrating each sampler into the same pipeline and how can they be resolved?

1.3 Outline

The remainder of this report is organized as follows.

Chapter 2 reviews the methodological background: Mathematics, Bayesian calibration, the all samplers under comparison, and the interfacing tools on which the benchmark is built.

Chapter 3 presents the rwmcmc (my implementation so it might be better to start with), other samplers' logics,the pipeline work, benchmark itself, sampler-agnostic definition of the posterior, the integration of each sampler into a common pipeline, and the implementation challenges encountered along the way, together with their solutions.

Chapter 4 compares the samplers against the research questions: consistency of the resulting posteriors, convergence diagnostics, and computational cost.

Chapter 5 summarizes the findings, discusses the limitations of the study, and outlines directions for future work.

2 Background and Related Work

Short chapter opening: what this chapter covers and in which order (inverse problem, Bayesian formulation, sampling, pipeline), one paragraph, written last.

2.1 The Inverse Problem

What it is, why it is important, when do we encounter it, and what are the main challenges. From measurements to parameters (noise, ill-posedness, non-uniqueness, etc.). What can be done? (I will not answer the question here.)

2.2 The Bayesian Approach to the Inverse Problem

From Bayes' theorem to prior, likelihood, and posterior (the details will be given in the chapter 3-Mathematical Formulations/Framework).

2.3 Sampling Methods for Bayesian Inference

Need for sampling, different samplers we implemented, each of them explained in short, their advantages and disadvantages, and the state of the art in the literature (their papers).

2.4 The Benchmarking Pipeline

Short description of the benchmarking pipeline, its components, and how it is used to compare the samplers.
Real simulations are expensive > need for surrogate models (RobustGaSP) > sampler and model connection (on UM-Bridge) > Nextflow. Here I can give a scheme.

3 Benchmarking Sampling Methods

Short chapter opening: what we built and why a common pipeline in which all samplers target the identical posterior, so that differences in results can be attributed to the samplers themselves. One paragraph.

3.1 Design of the Calibration Framework

The sampler-agnostic posterior definition: Prior, LogPrior, LogLikelihood, LogPosterior classes shared by all samplers. Why this design: any sampler only needs a function that evaluates the (unnormalized) log-posterior pointwise, so the Bayesian model is defined once and reused. Supported prior distributions (uniform, normal, truncated normal) and optional noise calibration. Common output contract across samplers (mcmc_output.npz, trace.npy, corner plot, ArviZ InferenceData) so that downstream diagnostics are identical. Here I can give a scheme of the class structure / data flow (parameters in, log-posterior out; sampler on top).

3.2 Mathematical Framework

The formulas promised in Chapter 2: posterior proportional to likelihood times prior; Gaussian log-likelihood with (optionally calibrated) noise sigma; log-domain evaluation and why (numerical underflow). For dynesty: the unit-cube prior transform via the inverse CDF (ppf), and why nested sampling needs likelihood and prior separately instead of a combined log-posterior (avoiding double-counting the prior). Only equations that are referenced later.

3.3 Starting With the Basics: Random-Walk Metropolis Hastings (RWMH)

Here, RWMCMC is the only package I wrote so I'll briefly explain how I did it with the code blocks. Also, I will explain the MCMC logic from that algorithm, and how it is implemented in the code. I will also explain the tuning parameters and their effects on the sampling process.

3.4 Sampler Integration

One subsection per sampler: how each one plugs into the shared posterior, its main tuning parameters, and how its output is mapped to the common contract.

3.4.1 emcee

Here I will explain each sampler in detail (under different headings), how do they work, what is important about them. I will also explain the tuning parameters and their effects on the sampling process.

3.4.2 dynesty

done

3.4.3 flowMC

planned

3.4.4 cdream

planned

3.4.5 PyMC

planned

3.4.6 Pigeons.jl

planned

3.5 Implementation Challenges and Solutions

The practical difficulties encountered while integrating the samplers, and how each was resolved. Pickling and multiprocessing: closures are not picklable (no importable top-level name), so the prior transform was refactored into a top-level class with __call__, instantiated inside Prior, why class instances work where closures fail. Parallelization of dynesty via fork context and the nprocs parameter. The rpy2 / R-based surrogate constraint: the forward model itself cannot be parallelized (single worker). Environment pinning: dynesty from conda-forge, matplotlib pin due to arviz incompatibility. Each challenge: symptom, cause, solution, lesson learned.

4 Evaluation

Short chapter opening: the samplers are compared against the research questions from Chapter 1 - consistency of the posteriors (RQ1), computational cost (RQ2); the integration experience (RQ3) was covered in Section 3.5. One paragraph.

4.1 Experimental Setup

Everything needed to reproduce the comparison: the calibration data; the priors on mu and xi (and noise sigma if calibrated); the settings of each sampler (nwalkers/nsteps/burn-in for emcee, nchains/step size for rwmcmc, nlive/stopping criterion for dynesty) and how these were chosen to make the comparison fair; software versions and hardware; the fact that all samplers ran through the same Nextflow pipeline against the same surrogate.

4.2 Metrics

Which quantities are compared and why each is meaningful. Posterior agreement: means, credible intervals, corner plots across samplers. Convergence diagnostics: R-hat and effective sample size (ESS), what they measure, and the caveat that R-hat is undefined for nested sampling (no chains) - why ESS is the appropriate diagnostic for dynesty. Computational cost: wall-clock time and number of forward model evaluations, and why model evaluations matter more than raw time when the forward model dominates the budget.

4.3 Results

The measured outcomes, presented objectively. Posterior comparison: corner plots of all samplers side by side (or overlaid), a table of posterior means and credible intervals for mu and xi per sampler. Diagnostics table: R-hat (where defined) and

ESS per sampler, including the dynesty R-hat FAIL as a false negative with ESS confirming convergence. Timing table: wall-clock and model evaluations per sampler. Figures and tables to be copied from the mcmc-bench outputs at the end. Basically, the results file in the report.

4.4 Discussion

Interpretation of the results, mapped back to the research questions. RQ1: do the posteriors agree, and within what tolerance. RQ2: which sampler is cheapest per effective sample, and how the non-parallelizable forward model shapes this comparison. Trade-offs: tuning burden vs. robustness vs. extra outputs (evidence from nested sampling). Honest limitations of the comparison itself (single problem, single surrogate, settings not exhaustively tuned).

5 Summary and Future Work

Short chapter opening: one paragraph restating the problem (sampler choice for Bayesian calibration of the mass flow model) and what the report contributes.

5.1 Summary

Research question by research question: RQ1 - whether the samplers produced consistent posteriors for mu and xi (answer from Section 4.3); RQ2 - how the computational costs compared; RQ3 - the main implementation challenges (pickling, parallelization limits, environment pinning) and their solutions. Key contribution: a sampler-agnostic benchmarking pipeline that makes such comparisons repeatable.

5.2 Limitations

Boundaries of the study: a single calibration problem and a single (surrogate) forward model; results may not transfer to higher-dimensional posteriors; the surrogate replaces the true simulator, so calibration quality is bounded by surrogate quality; sampler settings chosen reasonably but not exhaustively optimized; the single-worker constraint of the R-based model shapes the cost comparison for parallel cases. Also, the study focuses on a specific type of Bayesian calibration problem, and the results may not be directly applicable to other types of problems.

5.3 Future Work

The pipeline is designed to grow: integration of the planned samplers (flowMC, cdream, PyMC, Pigeons.jl - Section 3.4), which broadens the comparison to flow-accelerated, differential-evolution, and parallel tempering methods. Beyond samplers: higher-dimensional calibration (more parameters, noise), alternative forward models, and lifting the single-worker constraint (e.g., multiple surrogate instances) to make parallel cost comparisons meaningful. Lastly, the surrogate model does not create gradients, so the pipeline cannot yet compare gradient-based samplers against the others. Future work could include a differentiable surrogate to enable such comparisons or we can add some methods like Algorithmic Differentiation (AD) to the existing surrogate to make it differentiable. This would allow for the integration of gradient-based samplers and provide a more comprehensive comparison of sampling methods for Bayesian calibration problems.

A Supplementary Material

Include here any material that supports the main text but would disrupt the flow of the thesis if placed inline. Typical appendix content includes raw data, additional figures, extended tables, questionnaires, or code listings.[1]

[1] Only include material that is genuinely referenced from the main text. An appendix is not a dump for unused work.

B Additional Figures and Tables

Place additional figures and tables here that were omitted from the main chapters for brevity but are referenced in the text.